SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1741**

CO1-AT2-Constraint-Based Problem Solving

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COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

Assessment Tool Weightage: 20%

QUESTIONS:3 **Total Marks:30**

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

   Constraints:

   i. D1 cannot work Night shift
   ii. D2 must work before D3 (Morning < Afternoon < Night)
   iii. Only one doctor per shift
   iv. D3 cannot work Morning
   v. All doctors must be assigned exactly one shift

   Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

   Constraints:

   i. Movement allowed: Up, Down, Left, Right
   ii. Cost of each move = 1
   iii. Cannot pass through obstacles
   iv. Use Manhattan Distance heuristic

   Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost**.**

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.
   The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

   Constraints:

   - The robot can move up, down, left, right

- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:
a) Analyze all the given constraints and explain how they affect the robot's decision-making process.
b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.
c) Demonstrate how the robot applies AI concepts such as:
- Intelligent agents
- Rationality
- Environment type

d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

Code of Conduct:

I _D.N Shirshika(192425396) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *D.N Shirshika*

RUBRICS FOR CALCULATION:

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Criteria	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Max Marks**
Problem Understanding	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly	6
Constraint Analysis	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization	6
Solution Development	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints	7.5
Application of Concepts	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts	6

Justification	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided	4.5

ANSWERS

1)Backtracking Search for Doctor Shift Assignment

1. Aim

To assign three doctors (D1, D2, D3) to three shifts (Morning, Afternoon, Night) using the Backtracking Search algorithm while satisfying all the given constraints.

2. Problem statement

A hospital needs to assign three doctors to three shifts. Each doctor must work exactly one shift, and each shift must have only one doctor. The assignment must satisfy the following constraints:

1. D1 cannot work the Night shift.
2. D2 must work before D3.
3. Only one doctor is allowed per shift.
4. D3 cannot work the Morning shift.
5. All doctors must be assigned exactly one shift.

3. Algorithm

1. Start with an empty assignment.
2. Select a doctor and assign an available shift.
3. Check whether the assignment satisfies the constraints.
4. If any constraint is violated, backtrack and try another shift.
5. Repeat until all doctors are assigned.
6. Display the final valid schedule.

4. Step-by-step execution

Step	Assignment	Constraint check	Result
1	D1 = Morning	D1 is not assigned Night	Valid
2	D2 = Afternoon	D2 is before D3	Valid
3	D3 = Night	D3 is not assigned Morning	Valid
4	All doctors assigned	Every shift has one doctor	Valid

Intermediate steps

Step 1: Assign D1 to Morning. The assignment satisfies the first constraint.

Step 2: Assign D2 to Afternoon. This leaves Night available for D3 and maintains the required order.

Step 3: Assign D3 to Night. D3 is not assigned Morning, and D2 works before D3.

Step 4: Verify all constraints. Every doctor has exactly one shift, and no two doctors share a shift.

5. Final valid schedule

Shift	Doctor
Morning	D1
Afternoon	D2
Night	D3

6. Constraint verification

Constraint	Status
D1 cannot work Night	Satisfied
D2 works before D3	Satisfied
One doctor per shift	Satisfied
D3 cannot work Morning	Satisfied

Every doctor gets one shift Satisfied

7. Advantages of Backtracking Search

1. It checks constraints at every assignment step.
2. It rejects invalid assignments early.
3. It systematically explores possible solutions.
4. It can find a valid schedule without checking every possible assignment.

8. Result

The Backtracking Search algorithm successfully assigns D1 to Morning, D2 to Afternoon, and D3 to Night. All constraints are satisfied, so the final schedule is valid. Since the first attempted sequence is valid, no backtracking is needed in this execution**.**

2) BFS Search for a Robot in a 5×5 Grid

1.Aim

To find the shortest path from Start (S) to Goal (G) using Breadth-First Search (BFS) while avoiding obstacles.

2. Problem statement

A robot moves in a 5×5 grid using Up, Down, Left, and Right movements. Each move costs 1, and obstacles cannot be crossed. Manhattan distance is used to estimate the distance to the goal.

3. Constraints

1. Grid size is 5×5.
2. Movement is allowed in four directions.
3. Each movement costs 1.
4. Obstacles cannot be crossed.
5. The shortest path must be found.

4. Algorithm

1. Insert the start node into the queue.
2. Remove the first node from the queue.
3. Check its neighboring cells.
4. Ignore obstacles and visited cells.
5. Add valid neighbors to the queue.
6. Repeat until the goal is reached.
7. Trace the parent nodes to obtain the shortest path.

5. Step-by-step node expansion

Step	Expanded node	New nodes added to queue
1	(1,1)	(1,2), (2,1)
2	(1,2)	(1,3)
3	(2,1)	(3,1)
4	(1,3)	None
5	(3,1)	(4,1)
6	(4,1)	(4,2), (5,1)
7	(4,2)	(4,3), (5,2)
8	(5,1)	None
9	(4,3)	(3,3), (5,3)
10	(5,2)	None
11	(3,3)	(3,4), (2,3)
12	(5,3)	(5,4)
13	(3,4)	(3,5)
14	(2,3)	None

15	(5,4)	(5,5), Goal

6. Manhattan distance

The formula is:

$$ h(n)=|x_n-x_G|+|y_n-y_G| $$

For the start position (1,1) and goal (5,5):

$$ h(1,1)=|1-5|+|1-5|=8 $$

7. Optimal path

Start → (2,1) → (3,1) → (4,1) → (4,2) → (4,3) → (5,3) → (5,4) → Goal

Total cost:

$$ 8\times1=\boxed{8} $$

8. Result

BFS finds the shortest path from Start to Goal by expanding nodes level by level and avoiding obstacles. The robot reaches the goal in 8 moves with a total cost of 8.

3)Autonomous Rescue Robot Using AI Search Strategies

a) Analysis of Constraints

- Movement: The robot can move up, down, left, or right.
- Obstacles: Blocked cells must be avoided.
- Limited sensing: The robot can observe only adjacent cells.
- Movement cost: Each normal move costs 1.
- Risky zones: Entering a risky zone costs \(1+2=3\).
- Dynamic knowledge: The robot updates its map as it discovers new information.
- Safety: It must minimize cost while avoiding dangerous paths.

b) Solution Using UCS and Online Search

An Online Search Agent combined with Uniform Cost Search (UCS) is suitable.

Steps:

- Start at S and observe nearby cells.
- Identify obstacles, safe paths, and risky zones.
- Assign cost 1 to normal moves and cost 3 to risky moves.
- Use UCS to select the lowest-cost known route.
- Move one step and observe the surroundings.
- Update the map and replan if new obstacles are discovered.

[text{Total Cost}=\text{Normal Moves}+3(\text{Risky Moves})]

c) Application of AI Concepts

1. Intelligent Agent: The robot uses sensors to observe its surroundings and takes actions to reach the survivor.

2. Rationality: It selects the safest, lowest-cost route based on the information available and changes its plan when necessary.

3. Environment Type:

- Partially observable: Only adjacent cells are visible.
- Sequential: Each movement affects future decisions.
- Dynamic: The robot updates its knowledge during exploration.
- Discrete: Movement occurs between grid cells.
- Uncertain: Hidden obstacles and changing conditions may affect navigation.

d) Why Is This Approach Suitable?

UCS finds the least-cost route by considering normal movement costs and additional risk costs. Online search allows the robot to explore an unknown environment, update its map, avoid obstacles, and replan when new information becomes available.

Conclusion

An online search agent combined with UCS helps the rescue robot reach the survivor with minimum cumulative cost while adapting to unknown obstacles and prioritizing safe navigation.